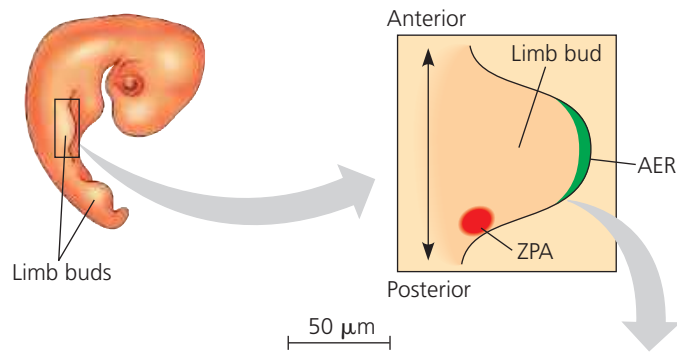
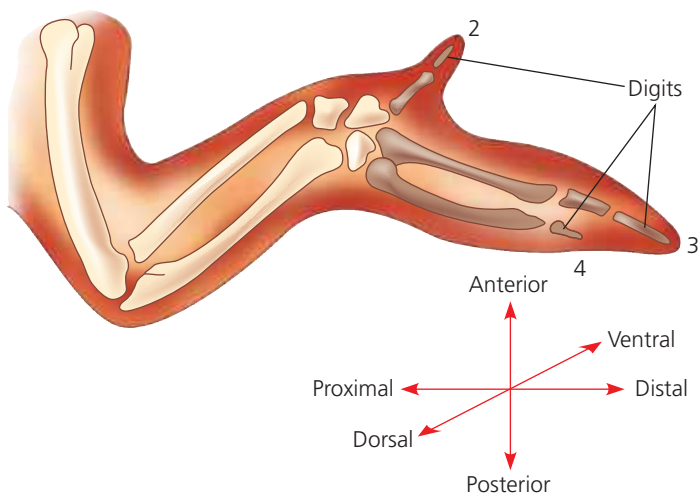
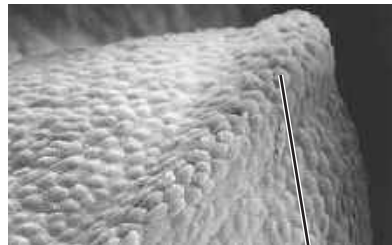


▼ **Figure 47.25 Regulation of vertebrate limb development by organizer regions.**



(a) Organizer regions.

Vertebrate limbs develop from protrusions called limb buds. Two regions in each limb bud, the apical ectodermal ridge (AER, shown in this SEM) and the zone of polarizing activity (ZPA), play key roles as organizers in limb pattern formation.



(b) Wing of chick embryo. Each embryonic cell receives positional information indicating location along the three axes of the limb. The AER and ZPA secrete molecules that help provide this information. (Numbers are assigned to the digits based on a convention established for vertebrate limbs. The chicken wing has only four digits; the first digit points backward and is not shown.)

Cells nearest the ZPA form posterior structures, such as the most posterior of the chick's digits (equivalent to our little finger); cells farthest from the ZPA form anterior structures, including the most anterior digit (like our thumb).

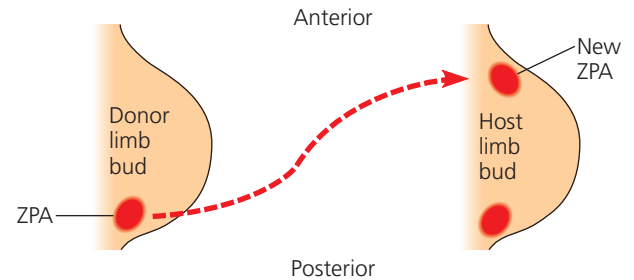
Like the AER, the ZPA influences development by secreting a protein signal. The signal secreted by the ZPA is called Sonic hedgehog, named after both a video game character and a similar protein in *Drosophila* that also regulates development. Implanting cells genetically engineered

▼ **Figure 47.26 Inquiry**

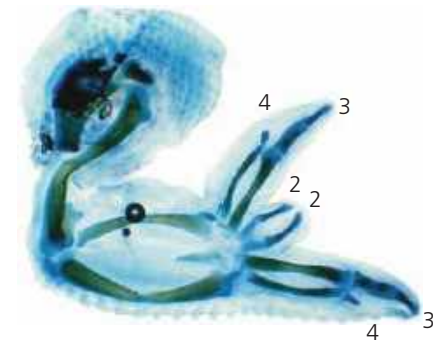
What role does the zone of polarizing activity (ZPA) play in limb pattern formation in vertebrates?



Experiment In 1985, researchers were eager to investigate the nature of the zone of polarizing activity. They transplanted ZPA tissue from a donor chick embryo under the ectoderm in the anterior margin of a limb bud in another chick (the host).



Results The host limb bud developed extra digits from host tissue in a mirror-image arrangement to the normal digits, which also formed (compare with Figure 47.25b, which shows a normal chick wing).



Conclusion The mirror-image duplication observed in this experiment suggests that ZPA cells secrete a signal that diffuses from its source and conveys positional information indicating "posterior." As the distance from the ZPA increases, the signal concentration decreases, and hence more anterior digits develop.

Data from L. S. Honig and D. Summerbell, Maps of strength of positional signaling activity in the developing chick wing bud, *Journal of Embryology and Experimental Morphology* 87:163–174 (1985).

WHAT IF? Suppose you learned that the ZPA forms after the AER, leading you to develop the hypothesis that the AER is necessary for formation of the ZPA. If you removed the AER and looked for expression of *Sonic hedgehog*, how would that test your hypothesis?

to produce Sonic hedgehog into the anterior region of a normal limb bud causes formation of a mirror-image limb—just as if a ZPA had been grafted there (**Figure 47.26**). Furthermore, experiments with mice reveal that production of Sonic hedgehog in part of the limb bud where it is normally absent can result in extra toes.

The AER and ZPA regulate the axes of a limb bud, but what determines whether the bud develops into a forelimb or hind